Gwynedd Mercy University-School of Business & Education
(*Must be used for each lesson observed by University Supervisor*)
Lesson Plan Framework

Subject Area: Learning Support Math **Grade Level: 9**

Student Name: Grace Dittman

Domain I: Planning and Preparation	
PA Standard(s): • CC.2.2.HS.D.9 / CC.2.2.7.EE.A.1 (foundational skill per IEP goals): Evaluate algebraic expressions by substituting given values for variables and applying order of operations.	
Essential Question(s): Students will consider and reflect on throughout the lesson: • How do we evaluate an algebraic expression when variables are replaced with given values? • How does order of operations guide us in evaluating expressions accurately?	
Objectives/Performance Expectations What will students know and be able to do as a result of this lesson? • Students will be able to order a set of numbers, including decimals and square roots, from largest to smallest. • Students will be able to substitute given values for variables in an algebraic expression, using parentheses around each substituted value. • Students will be able to apply order of operations to correctly evaluate an expression after substitution. • Students will be able to evaluate expressions involving exponents, absolute value, and multiple variables.	
Assessment Evidence/Level of Learning Students will show what they know by evidence of/list names of formative or summative checks, quizzes, tests which indicate evidence of learning: • Table-group ordering warm-up (checked together on the board) • Guided notes examples completed together	**Differentiated Instruction**: Modifications in instruction and assessment for students with learning differences, specifically those with 504's and IEP's. Extensions/Enrichment in instruction and assessment for students who are above grade level and/or gifted.

• Independent notes practice problems (#3-12 and summary) • Completed practice worksheet on evaluating expressions • Completion of self-selected choice activity (color-by-number or mixed review worksheet)	• Visual schedule posted and referenced to support transitions and predictability • Notes page chunked; work through examples together before releasing to independent practice • Reminder/anchor chart for order of operations (PEMDAS) available for reference • Substitution property reinforced concretely: 'put parentheses around the number every time you substitute' • Calculator use permitted as needed per IEP accommodations • Extended time provided for independent and practice worksheet portions • Student choice of closing activity (color-by-number vs. mixed practice) to support engagement and autonomy • Small-group/table support during warm-up and practice worksheet

Domain II: Classroom Environment:

Describe Classroom/Behavior Management Plan & Routines

• Visual schedule displayed at the start of class showing today's materials and sequence of activities, including the lunch break

• Warm-up completed in table groups with an 8-minute visible timer to build routine and time-awareness

• Clear verbal and visual cues signal transitions, especially around the lunch break

• Consistent routine: warm-up → review → notes/guided practice → independent practice → lunch → review → practice worksheet → choice activity → closure

• Positive reinforcement for on-task effort during independent work and choice activity

Domain III: Instruction

Activate Prior Knowledge/Motivation:

Preparing students for the lesson: How will you activate prior knowledge, build background, or review previous lessons as well as motivate students for the lesson?

• (10:47-10:50) Students enter, review the visual schedule on the board showing materials needed and today's activities.
• (10:50-10:58) Warm-up: In table groups, students receive small cards with numbers (decimals, square roots, etc.) and order them from largest to smallest. An 8-minute timer is displayed.
• (10:58-11:05) Re
view the warm-up together as a class on the board, discussing strategies for comparing decimals and square roots.

Materials Needed:
• Visual schedule (board/projector)
• Warm-up number-ordering cards (decimals, square roots, etc.) — one set per table group
• Visible timer
• Evaluating Expressions guided notes (All Things Algebra)
• Whiteboard/projector for modeling examples
• Independent practice worksheet (additional evaluating expressions problems)
• Choice activity: color-by-number order of operations worksheet OR mixed practice worksheet (evaluating expressions, absolute value, ordering numbers, order of operations)
• Pencils, calculators (per accommodations)

Vocabulary (New or Review):
• Variable — a letter that represents an unknown or changing number
• Substitution — replacing a variable with a given number
• Evaluate — to find the value of an expression
• Order of operations (PEMDAS) — the order in which operations must be performed
• Exponent — review
• Absolute value — review

Sequence of the Lesson:
What learning activities/strategies will you use to engage the students in the learning? What will students do to use and apply new concepts or skills (independent practice if relevant)? How will you monitor and guide their performance? Include relevant vocabulary. Use bullets to sequence your lesson.

• (10:47-10:50) Students enter, review the visual schedule on the board showing materials needed and today's activities.
• (10:50-10:58) Warm-up: In table groups, students receive small cards with numbers (decimals, square roots, etc.) and order them from largest to smallest. An 8-minute timer is displayed.
• (10:58-11:05) Re
view the warm-up together as a class on the board, discussing strategies for comparing decimals and square roots.
• (11:05-11:11) Pass out Evaluating Expressions notes. Introduce the substitution property: "If a variable is replaced with a number, we substitute that value in for the variable." Emphasize: "Each time you substitute a variable with a number, put parentheses around the number!" Review that students must follow order of operations to evaluate. Model example 1 (ab^2 + c).

• (11:11-11:17) Model example 2 (3x^2 - 4x), thinking aloud through substitution, parentheses, and order of operations at each step.
• (11:17-11:23) Guided practice: work through 'You Try' problems 3-4 together as a class, checking for understanding.
• (11:23-11:29) Students begin independent practice on problems 5 and beyond; work will continue after lunch.
• --- LUNCH BREAK ---
• (upon return, ~5 min) Students return, review visual schedule to re-orient to the remainder of class.
• (12:05-12:20) Students finish independent practice on the notes page (problems 5-12 and summary); review answers together as a class, addressing common errors.
• (12:20-12:25) Pass out and explain the new practice worksheet with additional evaluating expressions problems.
• (12:25-12:46) Students work independently (or with table support) on the practice worksheet.
• (remaining time) Choice activity: students select either a color-by-number order of operations activity or a mixed practice worksheet covering evaluating expressions, absolute value, ordering numbers, and order of operations, based on preference and need.

Closure: What will you do to bring closure to the lesson? How will you summarize this lesson and preview the lesson that will follow?
• Bring the class back together and review 1-2 problems as a quick exit check.
• Recap the big idea: substitute carefully with parentheses, then follow order of operations to evaluate.
• Preview that the next lesson will continue building on evaluating expressions and/or order of operations.
• Collect notes, practice worksheet, and choice activity.

Reflection Guidelines

The student will write lesson plans with an eye to the Danielson Framework and the Four Domains of Professional Practice. Use this as a resource when writing lesson plans.

Domain I Planning and Preparation

1. How do you know you were knowledgeable of your content?
2. What is the evidence that your objective was clear in the lesson plan?
3. How does the lesson address individual student learning needs?
4. What variety of strategies and methodologies did you consider when planning?
5. What is the evidence that lesson components were logically sequenced?
6. How did your lesson plan include checks for student understanding?
7. What PA Standards did this lesson address?

Domain II Classroom Environment

1. How did students know the behavioral and academic expectations for the lesson?
2. What effective routines and procedures did you implement for lesson efficiency?
3. How did you insure the classroom was physically accessible for all learners?
4. What evidence indicates that respect was shown between and among students and teacher?
5. What evidence indicates that students placed a high value on quality participation, process, and product?
6. How do you assess if the lesson was attainable for all students but also had appropriate rigor?

Domain III Instructional Delivery

1. How did you align your lesson to PA Standards?
2. How did you insure a clearly stated objective, understood by learners?
3. How did you provide clear explanations as to how to proceed through the lesson?
4. What questioning and discussion techniques insured all-learner participation?
5. What strategies did you employ to engage all students?
6. Describe how you made the content meaningful due to sequencing and pacing.
7. Discuss how you checked for understanding, provided feedback, retaught.
8. What is the evidence that you were you flexible and responsive to the needs of the learners?
9. Describe how you made meaning of the lesson with some form of closure activity.

Overall Assessment

Briefly state any changes you will make the next time you teach this lesson or activity, to improve any of the Domains of Professional Practice.